Sedimentation in a box

Answer key — Journal club

$\mathrm{d}x = -v\,\mathrm{d}t + \sqrt{D}\,\mathrm{d}W$ on $0 < x < L$, with $h = D/2v = k_BT/mg$, $k_n = n\pi/L$, and $B = D$ the variance per unit time. Closed cell: reflecting walls. Sticky cell: absorbing walls. Equation numbers are Gardiner, 2nd edition.

Key results

forward equation:	$\partial_t p = v\,\partial_x p + \frac{1}{2}D\,\partial_x^2 p = -\partial_x J, \qquad J = -vp - \frac{1}{2}D\,\partial_x p$
closed, stationary:	$p_s = \dfrac{e^{-x/h}}{h\,(1-e^{-L/h})}, \qquad \langle x\rangle_s = h - \dfrac{L}{e^{L/h}-1}$
both cells:	$\lambda_n = \dfrac{v^2}{2D} + \dfrac{n^2\pi^2 D}{2L^2}$ $(n \geq 1)$; the closed cell also has $\lambda_0 = 0$
closed modes:	$Q_n = e^{x/2h}\varphi_n, \quad P_n = \dfrac{e^{-x/2h}\varphi_n}{N_n}, \quad$ with $\quad \varphi_n = \cos k_n x - \dfrac{\sin k_n x}{2hk_n}$
normalisation:	$N_n = \displaystyle\int_0^L \varphi_n^2\,\mathrm{d}x = \frac{L}{2}\Big(1 + \frac{1}{4h^2k_n^2}\Big)$
sticky modes:	$Q_n = e^{x/2h}\sin k_n x, \quad P_n = \dfrac{2}{L}\,e^{-x/2h}\sin k_n x, \quad q = \dfrac{(2h)^{-2}+k_1^2}{k_1\,(1+e^{-L/2h})}\,e^{-x/2h}\sin k_1 x$
sticking:	$\pi_L = \dfrac{e^{x_0/h}-1}{e^{L/h}-1}, \qquad T = \dfrac{x_0 - L\,\pi_L(x_0)}{v}$
from the QSD:	$\langle T\rangle = \dfrac{1}{\lambda_1}, \qquad \Pr(\text{ceiling}) = \dfrac{1}{1+e^{L/2h}}$
the map:	$J\big[P_n^{\text{closed}}\big] = \frac{1}{2}Dk_n\,P_n^{\text{sticky}}, \qquad P_n^{\text{closed}} = \partial_x J\big[P_n^{\text{closed}}\big]/\lambda_n$

(a) The closed cell: find the stationary state

1. $A = -v$, $B = D$, so $\partial_t p = v\,\partial_x p + \frac{1}{2}D\,\partial_x^2 p = -\partial_x J$ with $J = -vp - \frac{1}{2}D\,\partial_x p$. Hard walls reflect: $J(0,t) = J(L,t) = 0$.

2. $2A/B = -2v/D = -1/h$, so (5.2.39) gives $p_s \propto e^{-x/h}$, and

$$\boxed{p_s(x) = \frac{e^{-x/h}}{h\,(1-e^{-L/h})}, \qquad h = \frac{D}{2v} = \frac{2k_BT/\gamma}{2mg/\gamma} = \frac{k_BT}{mg}}$$

This is the Boltzmann factor e^{-mgx/k_BT}. Friction sets how fast the particle moves, not where it ends up in equilibrium, so γ cancels between v and D.

3. $m = \frac{4}{3}\pi r^3\Delta\rho = 5.2\times10^{-16}\,\mathrm{kg}$, $mg = 5.1\times10^{-15}\,\mathrm{N}$, so $h \approx 0.8\,\mu\mathrm{m}$: the colloid's atmosphere is a micron thick, and the cell is tall on that scale. With $\gamma = 6\pi\eta r$ and $\eta = 1.0\,\mathrm{mPa\,s}$, the settling speed is $v \approx 0.55\,\mu\mathrm{m/s}$, so $h/v \approx 1.4\,\mathrm{s}$.

4. $\langle x\rangle_s = \int_0^L x\,p_s\,\mathrm{d}x = h - \dfrac{L}{e^{L/h}-1}$. For $L \ll h$, expanding gives $\langle x\rangle_s \simeq \frac{L}{2} - \frac{L^2}{12h}$: nearly uniform. For $L \gg h$, $\langle x\rangle_s \to h$: the floor's atmosphere, which does not feel the ceiling.

(b) The closed cell: find the modes

1. e^{ax} needs $\frac{1}{2}Da^2 - va + \lambda = 0$, so $a = [v \pm \sqrt{v^2 - 2D\lambda}\,]/D = 1/2h \pm ik$ with $\lambda = v^2/2D + \frac{1}{2}Dk^2$.

Take $Q = e^{x/2h}(\cos kx + c\sin kx)$. $Q'(0) = 0$ gives $c = -1/2hk$, and then

$$Q'(L) = -e^{L/2h}\Big(k + \frac{1}{4h^2k}\Big)\sin kL = 0 \quad\Longrightarrow\quad k_n = \frac{n\pi}{L}.$$

For real a, put $k = i\kappa$: the condition becomes $(\kappa - 1/4h^2\kappa)\sinh\kappa L = 0$, so $\kappa = 1/2h$. That is $\lambda = 0$ with $Q = e^{x/2h}(\cosh\frac{x}{2h} - \sinh\frac{x}{2h}) = 1$, the stationary mode. There are no others.

2. With $\int p_s Q_n Q_m\,\mathrm{d}x = \delta_{nm}N_n/[h(1-e^{-L/h})]$, the normalised pair is

$$Q_n = e^{x/2h}\varphi_n, \qquad P_n = \frac{e^{-x/2h}\varphi_n}{N_n}, \qquad \varphi_n = \cos k_n x - \frac{\sin k_n x}{2hk_n}, \qquad N_n = \frac{L}{2}\Big(1 + \frac{1}{4h^2k_n^2}\Big).$$

Since $J[p_s] = 0$, $J[p_sQ] = -\frac{1}{2}D\,p_s\,Q'$, which vanishes wherever $Q' = 0$. The backward condition (5.2.69) is the forward one in disguise.

3. $\lambda_1 = v^2/2D + \pi^2 D/2L^2$. The second term is free diffusion across the cell, the gap of the talk's box. The first is added to every mode by the drift, whatever L: it is $v/4h$, the rate of settling through a few gravitational lengths. For $L \gg h$ it dominates, and $1/\lambda_1 \simeq 4h/v$, independent of L.

4. At $L = 20h$: $\lambda_1 = (0.25 + 0.025)\,v/h$, so $1/\lambda_1 \approx 3.6\,h/v$, while settling from the ceiling takes about $20\,h/v$. The expansion explains it:

$$p(x,t\,|\,x_0,0) - p_s(x) = e^{(x_0-x)/2h}\sum_{n\ge1}\frac{\varphi_n(x)\,\varphi_n(x_0)}{N_n}\,e^{-\lambda_n t}.$$

For $x_0 = L$ and x near the floor, the prefactor is $e^{L/2h} = e^{10} \approx 2\times10^4$, while $p_s \sim 1/h$. The modes cancel almost exactly at first, which is how the density near the floor starts at zero. The deviation is small only once $e^{L/2h}e^{-\lambda_1 t} \ll 1$, that is, after $t \approx (L/2h)/\lambda_1 \approx 2L/v$: of the order of the settling time. The gap sets the final exponential approach, not when it begins. Large coefficients like these are the price of a non-symmetric operator. In the symmetrised variables $e^{\pm x/2h}$ is removed, and nothing is paradoxical.

(c) Make the walls sticky

1. A stationary state has constant J, so $-vp - \frac{1}{2}Dp' = J$, with solution $p = Ce^{-x/h} - (J/v)(1-e^{-x/h})$. $p(0) = 0$ gives $C = 0$, and $p(L) = 0$ then gives $J = 0$, so $p = 0$. Every particle eventually sticks.

2. $Q = e^{x/2h}(\alpha\cos kx + \beta\sin kx)$ with $Q(0) = 0$ gives $\alpha = 0$. $Q(L) = 0$ gives $\sin kL = 0$, so $k_n = n\pi/L$ again. Real a fails, because $\sinh\kappa L \neq 0$. So

$$\boxed{Q_n = e^{x/2h}\sin k_n x, \qquad \lambda_n = \frac{v^2}{2D} + \frac{n^2\pi^2 D}{2L^2}, \qquad n \ge 1}$$

exactly the nonzero rates of the closed cell. Only $\lambda_0 = 0$ is lost (panel c). Part (e) explains why.

3. $p_sQ_n \propto e^{-x/2h}\sin k_n x$. Trying e^{ax} in the forward equation $vP' + \frac{1}{2}DP'' = -\lambda P$ gives $a = -1/2h \pm ik$ with the same λ, and $p = 0$ at the walls holds. Gardiner's (5.2.91) keeps the reflecting p_s for absorbing walls. Since $\int p_sQ_nQ_m\,\mathrm{d}x \propto \int_0^L \sin k_n x \sin k_m x\,\mathrm{d}x$, the normalised pair is $Q_n = e^{x/2h}\sin k_n x$ and $P_n = \frac{2}{L}\,e^{-x/2h}\sin k_n x$.

4. $q = P_1/\int P_1\,\mathrm{d}x$. With the integral in the hint,

$$\boxed{q(x) = \frac{(2h)^{-2} + k_1^2}{k_1\,(1 + e^{-L/2h})}\;e^{-x/2h}\sin\frac{\pi x}{L}}$$

Its peak is where $\tan(\pi x^*/L) = 2\pi h/L$: $x^* \approx 1.7h$ for $L = 8h$, and $x^* \to 2h$ for $L \gg h$ (panel a). Why higher than p_s: $q \propto p_s Q_1$, and $Q_1(x_0)$ is the long-time survival weight, $S(t\,|\,x_0) \simeq Q_1(x_0) \int P_1\,\mathrm{d}x\; e^{-\lambda_1 t}$. Particles low down soon touch the floor, so the survivors come from higher up, and the weight $e^{x/2h}$ turns $e^{-x/h}$ into $e^{-x/2h}$.

5. The approach rate is $\lambda_2 - \lambda_1 = 3\pi^2 D/2L^2$. The drift adds the same $v^2/2D$ to every mode, so it cancels in every difference. Gravity speeds up sticking, but the survivors' shape settles by diffusion across the whole cell. For $L \gg h$, $\lambda_2 - \lambda_1 \ll \lambda_1$: most particles stick before the survivors' shape settles, the opposite of metastability. Panel (b) is at $L = 8h$, where the two rates are comparable.

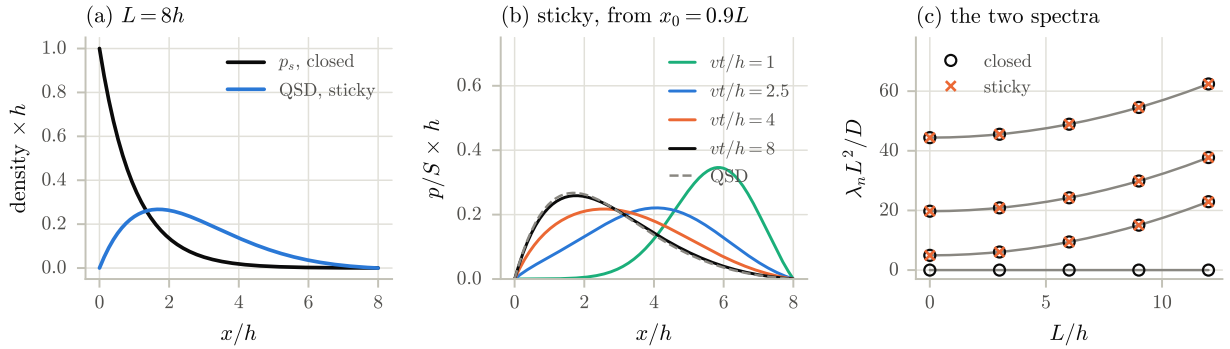


Figure 1: (a) Barometric p_s of the closed cell and the QSD of the sticky cell, $L = 8h$. (b) Sticky cell, $L = 8h$, particles released at $x_0 = 0.9L$: the density conditioned on survival, from the mode sum, settles onto the QSD. (c) Rates against L/h: curves are λ_n for $n = 0, \ldots, 3$; markers are eigenvalues of finite-difference matrices for the closed cell (circles) and the sticky cell (crosses).

(d) Where and when does it stick?

1. With $\psi = e^{-x/h}$, the corrected (5.2.189)–(5.2.190) give $\pi_L = \int_0^{x_0} e^{y/h}\mathrm{d}y / \int_0^L e^{y/h}\mathrm{d}y = (e^{x_0/h} - 1)/(e^{L/h} - 1)$. It solves $-v\pi' + \frac{1}{2}D\pi'' = 0$, $\pi(0) = 0$, $\pi(L) = 1$, as (5.2.186)–(5.2.187) require. For $L \gg h$, $\pi_L \simeq e^{-(L-x_0)/h}$: only particles that start within a few h of the ceiling have a fair chance. It is the barometric factor again.

2. The general solution of $-vT' + \frac{1}{2}DT'' = -1$ is $T = x/v + c_1 + c_2 e^{x/h}$, and $T(0) = T(L) = 0$ give

$$\boxed{T(x_0) = \frac{x_0 - L\,\pi_L(x_0)}{v}}$$

For $L \gg h$, away from the ceiling, $\pi_L \approx 0$ and $T \approx x_0/v$: the time to fall to the floor.

3. Started as the QSD, $p(x,t) = q(x)\,e^{-\lambda_1 t}$ exactly, so $S(t) = e^{-\lambda_1 t}$: the time to stick is exponential with mean $1/\lambda_1$. At the walls $q = 0$, so $J = -\frac{1}{2}Dq'$. Writing $q = C\,e^{-x/2h}\sin k_1 x$,

$$J(0) = -\tfrac{1}{2}DCk_1 \quad \text{(down, out through the floor)}, \qquad J(L) = +\tfrac{1}{2}DCk_1\,e^{-L/2h}.$$

The total outflux is $\frac{1}{2}DCk_1(1 + e^{-L/2h}) = \frac{1}{2}D\big[(2h)^{-2} + k_1^2\big] = \lambda_1$. The ceiling's share is $e^{-L/2h}/(1 + e^{-L/2h}) = 1/(1 + e^{L/2h})$ at every moment, and so overall: 1.8% for $L = 8h$.

4. Averaging over starting points drawn from q:

$$\int_0^L q\,T\,\mathrm{d}x = \frac{1}{\lambda_1}, \qquad \int_0^L q\,\pi_L\,\mathrm{d}x = \frac{1}{1 + e^{L/2h}}.$$

Both are hard integrals made easy (the script checks them).

(e) Optional: why the spectrum survived

1. With constant A and D, $\partial_t J = A\,\partial_t p - \frac{1}{2}D\,\partial_x\partial_t p = -A\,\partial_x J + \frac{1}{2}D\,\partial_x^2 J$: the same forward equation.

2. In the closed cell $J = 0$ at both walls at all times, so J is a solution of the sticky-cell problem. A mode $P_n e^{-\lambda_n t}$ has current $J[P_n]e^{-\lambda_n t}$, a sticky mode with the same λ_n, unless the current vanishes. Explicitly, with $\varphi_n' = -k_n \sin k_n x - \frac{1}{2h}\cos k_n x$,

$$J[P_n^{\text{closed}}] = -\frac{D}{2N_n}\,e^{-x/2h}\Big(\frac{\varphi_n}{2h} + \varphi_n'\Big) = \frac{D}{2N_n}\Big(k_n + \frac{1}{4h^2 k_n}\Big)e^{-x/2h}\sin k_n x = \tfrac{1}{2}Dk_n\,P_n^{\text{sticky}}.$$

The stationary mode has $J[p_s] \equiv 0$: its current is not a mode, and so λ_0 has no partner.

3. Let $\mathcal{L}\tilde{p} = -A\tilde{p}' + \frac{1}{2}D\tilde{p}'' = -\lambda\tilde{p}$ with $\tilde{p} = 0$ at the walls, and set $P = \tilde{p}'/\lambda$. Then $J[P] = (A\tilde{p}' - \frac{1}{2}D\tilde{p}'')/\lambda = -\mathcal{L}\tilde{p}/\lambda = \tilde{p}$, which vanishes at the walls, and $\mathcal{L}P = -\partial_x J[P] = -\tilde{p}' = -\lambda P$. So P is a closed-cell mode. The two maps are inverse to each other, and the spectra agree except for λ_0. In particular $P_n^{\text{closed}} = \partial_x J[P_n^{\text{closed}}]/\lambda_n$.

4. Constant A was used to move A through ∂_x. With $A = -U'(x)$ and constant D, $\partial_t J = -A\,\partial_x J + \frac{1}{2}D\,\partial_x^2 J$. That is the *backward* equation of the reflected potential $-U$, with $J = 0$ at the walls. So the closed problem in U and the sticky problem in $-U$ share every nonzero rate. These are the supersymmetric partners of quantum mechanics, here in imaginary time. For constant drift, reflecting the potential changes nothing, since λ_n depends only on v^2.

For the discussion

- **Survivors feel no gravity.** Weighting paths by their chance of surviving forever (Doob's h-transform, with $h = Q_1$) adds the drift $D\,\partial_x \ln Q_1 = v + (\pi D/L)\cot(\pi x/L)$. The particle conditioned never to stick has drift $(\pi D/L)\cot(\pi x/L)$, whatever v, and stationary density $P_1 Q_1 \propto \sin^2(\pi x/L)$, symmetric. The QSD is lopsided because it describes particles that have survived *so far*, not forever.
- **Same spectrum, different physics.** In the closed cell, λ_1 is the rate of approach to equilibrium. In the sticky cell, the same number is the rate of loss.
- **Checks.** `scripts/sedimentation.py` checks p_s, $\langle x\rangle_s$ and the silica numbers; the residuals, boundary conditions and biorthogonality of both sets of modes; both spectra against finite-difference matrices; the QSD, its peak and its currents; survival against a finite-difference solution; π_L, T and the two averages in (d); the map in (e); and the conditioned drift above, at $L/h = 2$, 10 and 0.33.